

AN EFFICIENT ALGORITHM FOR PASSIVE DOA TRACKING USING UNSCENTED KALMAN FILTER

C.VIJAY KUMAR

E.C.E. Dept., R.E. C
Tiruchirappalli - 620 015
Ph: 0431 - 500727
Cvkumar@rect.ernet.in

**Dr. R.RAJAGOPAL
AND N.RAVI KIRAN**

C.R.L., BEL
Bangalore -13
Ph:080 -8385905
rr@crlbel.ernet.in
nallaravi@yahoo.com

C.CHANDRASEKARAN

Hello Soft Inc
Hyderabad
Ph:040- 6502272
c_chandrasekaran@yahoo.com

ABSTRACT

Tracking of multiple targets by using only the bearing information is an important problem in passive sensing scenarios such as passive radar or infrared seeker applications. In this paper, a new algorithm is presented to adaptively estimate the AOAs of multiple moving targets when Linear Equispace Array (LES) is used for making the measurements. This algorithm is based on an extension of Differential MUSIC method. It can also discriminate sources and their multi-paths. The results obtained by this algorithm are then applied as input to a Non - Linear State Linear Measurements Unscented Kalman filter. The Unscented Kalman filter also adapts to maneuver. This algorithm gives 50percent reduction in the computation and memory requirements over unscented filter based tracker. Simulation results show the good performance of the filter over Unscented filter based tracker.

1. INTRODUCTION

Direction of Arrival (DOA) estimation in a passive radar scenario has attracted immense interest among researchers in the last two decades. Among the high resolution DOA estimation methods proposed, those based on the estimation of noise subspace / signal subspace vectors, in order to finally estimate the DOAs of the sources become very popular owing to their superior resolution capabilities and better bias performances. Notable among the Eigen structure methods is the MUSIC algorithm proposed by Schmidt [1]. This algorithm assumes that the source signals are at the most partially correlated and additive noise is spatially white.

In realistic passive radar scenario one may encounter multiple sources, which are highly or fully coherent, and an additive noise field corrupting the signal, which is directional and spatially correlated. The Differential Music Algorithm proposed by Rajagopal and Rao [2] can be applied to estimate the DOAs of these sources. This method has been shown to enjoy many advantages over conventional MUSIC. Using Differential MUSIC DOAs of correlated multipath arrivals could be estimated, but the source and its specular arrivals cannot be distinguished. M.Zoltowski and Haber extended the conventional MUSIC algorithm to discriminate between the direct path

arrival and specular path arrival. It can give faithful results only up to -5 dB. But the extended Differential MUSIC algorithm proposed here can work up to -20 dB.

A major disadvantage of eigen structure methods is their computational load. In order to reduce the computational complexity, Q-R decomposition of the difference matrix was applied in [3]. However this method is also a 'block' processing method. In order to make the DOA estimation algorithms adaptive, Yang and Kaveh [4], proposed updating techniques. These algorithms have still too high computational complexity and are not easily mapped into VLSI architectures (like systolic arrays) for real time applications. Hwang and Wei [5] have proposed a rank revealing QR updating procedure for DOA tracking. Though this method reduces the computational load considerably it cannot be applied to the generalized passive Sonar/Radar environment mentioned above. Rajagopal and Vijay kumar [6] have developed a dynamic and adaptive DOA estimation algorithm by using a Q-R updating procedure as and when new data arrive. Passive tracking of targets using angle only information is a classical navigation problem and has been extensively studied for naval applications [8-13] and for passive homing missile application [14]. Major efforts associated with the naval applications have been directed to analyze the observability of the tracking system and to find the proper filter structures for target state estimation. In all these works, the bearing information is assumed to be available through measurements made by some means. No effort has been made to tackle the measurement problem along with the tracking problem. KVS babu et.al. [15,16] and R. Rajagopal and Vijay Kumar Chakka [6] addressed this problem, but it is not possible to discriminate sources and their multi paths and also it is not possible for tracking of maneuverable targets. C.Vijaykumar and Rajagopal proposed [20] a unified procedure, which combines the DOA estimation and tracking using Unscented filter based tracker. This algorithm has high computational and memory requirements. This paper presents a computationally efficient algorithm. This Unscented Kalman filter algorithm utilises 50percent less data memory space and 50 percent reduction in the computation without reduction in performance. For adaptive DOA estimation, the Differential MUSIC algorithm is used. The results obtained are then applied as input to a Non-Linear State Linear Measurement Unscented Kalman filter. This filter avoids the derivation of the Jacobian matrices for linearizing the nonlinear kinematics model. The Unscented Kalman filter based tracker uses normalized innovation square (NIS) criterion to detect maneuver and accounts for it by continuous process noise level adjustment. Simulation results show the good performance of the filter over the Unscented filter based tracker [20]

This paper is organized into seven sections. Section 2 describes adaptive DOA Estimation. Section 3 presents Non Linear state Linear measurement for DOA Tracking. Section 4 presents DOA tracking using Unscented Kalman filter and section 5 presents the simulation results and performance comparisons with Unscented filter based tracker [20] and finally conclusions are presented in section 6.

2. ADAPTIVE DOA ESTIMATION

Measurement Equation:

Consider an LES Array with M identical sensors with equi-distance of Δ . Since both target and array are in motion, the received signal will be Doppler shifted in frequency. The array moves along the x -axis at a constant velocity. The received data vector during the first measurement will be given by

$$P(t_0) = a(\theta(t_0))s(t_0) + n(t_0) \quad \dots (2.1)$$

$$a(\theta_0) = \begin{bmatrix} 1 e^{-j(\omega/c)\Delta \sin \theta(t_0)} & \dots & e^{-j(\omega/c)(M-1)\Delta \sin \theta(t_0)} \end{bmatrix}^T \quad \dots (2.2)$$

Where ' c ' is the speed of propagation ω is the received signal angular frequency and θ is the DOA of target with respect to the array normal. In the case of multiple moving targets, it is assumed that the trajectories of targets do not cross each other. The measurement equations in this case are given by

$$P(t) = A(\theta(t_0))s(t_0) + n(t_0) \quad \dots (2.3)$$

$$P(t_k) = A(\theta(t_k))\phi(t, t_k)s(t_k) + n(t_k) \quad \dots (2.4)$$

where $A(\theta(t))$ is the array matrix with the direction vectors $a(\theta(t))$ as its columns. $\phi(t, t_k)$ is a matrix of order d (d is the total number of targets) having non-zero elements, which contain the information about the relative phase shift as compared to reference time t_0 .

The autocovariance matrix of the received data vector is given by

$$R \triangleq \sum_k [P(t_k)P^*(t_k)] = ASA^* + Q \quad \dots (2.5)$$

where S is signal covariance matrix, Q is the unknown noise spatial covariance matrix and A is the array direction matrix.

The following steps are involved in estimating the DOAs by using Extended Differential MUSIC:

1. Compute the auto covariance matrix R from the data vector.
2. Compute the difference matrix $D = R - \bar{E}RE$, where E is the $M \times M$ Exchange matrix.
3. Perform the eigen - decomposition of the matrix D .
4. Project the linear combination of the DOA vectors of the direct and specular paths on to the noise subspace. Compute the two-dimensional spatial spectra $S(\theta_d, \theta_s)$ [19].
5. Project the peak on to both the direct and specular path axes to get the respective arrival angles.

The above algorithm can be applied when there are p coherent groups of signals each containing two fully or partially coherent signals. For a more generic case, the higher order-differencing scheme [3] can be applied. Computation of eigen vectors of D will involve $O(M^3)$ flops. This is reduced by resorting to an adaptive QR - updating algorithm which requires only $O(M^2)$ flops.

3. NON-LINEAR STATE AND LINEAR MEASUREMENT MODEL FOR DOA TRACKING

X and Y are rectangular co-ordinates

Range = $r = \sqrt{X^2 + Y^2}$,

U = x-velocity, V= y-velocity, T = sampling time period

n_x and n_y are x and y process noises respectively

Let $[X_r \ Y_r \ U_r \ V_r] = [X/r \ Y/r \ U/r \ V/r]$

$$\text{den}(k) = \sqrt{1 + (U_r^2(k) + V_r^2(k)) * T^2 + 2 * U_r(k) * X_r(k) * T + 2 * V_r(k) * Y_r(k) * T} \quad \dots(3.1)$$

$$\text{PAR}_1(k) = (X_r(k) + U_r(k) * T) / \text{den}(k)$$

$$\text{PAR}_2(k) = U_r(k) / \text{den}(k)$$

$$\text{PAR}_3(k) = (Y_r(k) + V_r(k) * T) / \text{den}(k)$$

$$\text{PAR}_4(k) = V_r(k) / \text{den}(k)$$

$$G(k) = \begin{bmatrix} .5T^2 & 0 \\ T & 0 \\ 0 & .5T^2 \\ 0 & T \end{bmatrix} \div \text{den}(k) \quad \dots(3.2)$$

The state model or state equation is

$$\begin{bmatrix} X_r(k+1) \\ U_r(k+1) \\ Y_r(k+1) \\ V_r(k+1) \end{bmatrix} = \begin{bmatrix} \text{PAR}_1 \\ \text{PAR}_2 \\ \text{PAR}_3 \\ \text{PAR}_4 \end{bmatrix} + G(k) \begin{bmatrix} n_x & 0 \\ 0 & n_y \end{bmatrix} \quad \dots(3.3)$$

where n_x and n_y are independent, zero mean x and y Gaussian process noises. They represent accelerations normalized by range.

The measurement vector at the input of filter is

$$\begin{bmatrix} Z_x \\ Z_y \end{bmatrix} = \begin{bmatrix} \cos(\theta + w_x) \\ \sin(\theta + w_y) \end{bmatrix} \quad \dots(3.4)$$

where

Z_x = noisy $\cos(\theta)$ measurement, Z_y = noisy $\sin(\theta)$ measurement

w_x and w_y are the corresponding zero mean white Gaussian measurement noises.

4. DOA TRACKING OF TARGET USING UNSCENTED KALMAN FILTERING ALGORITHM

4.1 Introduction

One of the most fundamental tasks in filtering and estimation is to calculate the statistics of a random variable that has undergone a transformation. When the process and observation model are linear, the Kalman filter provide exact closed form solution. However, when the system is nonlinear, the exact value of the statistics must be calculated by integrating the transformation across the full conditional density, which requires unbounded number of calculations. In [7], a method for propagating means and covariances through nonlinear transformations called "unscented transformation" is proposed. A set of weighted sigma points are deterministically chosen so that certain properties of these points (such as their first two moments) match those of prior distribution. Each point undergoes the nonlinear transformation and the properties of the transformed set are calculated. The symmetric sigma point [7] solution uses $2n+1$ points to match the mean and covariance of a n -dimensional random variable. With this set of points unscented transform guarantees the same performance as the truncated second order filter, with the same order of calculations as an Extended Kalman filter (EKF) but without the need to calculate any approximations or derivatives. However, in extremely large state space systems or in real time applications it is critical that both computational and storage requirements are minimized. To address this problem Julier [21] proposed the Unscented Kalman Filter (UKF). The basic components of this filter are "unscented transform with reduced sigma points" (i.e. simplex sigma points) and "scaled unscented transformation".

The scaled unscented transform preserves the first two moments but scales the effect of all higher order moments by an arbitrary scalar. This is achieved by applying the unscented transform to an auxiliary random variable Z that is related to X through

$$Z = f(X) = \frac{f[\bar{X} + \alpha(X - \bar{X})] - f(\bar{X})}{\mu} + f(\bar{X}) \quad \dots(4.1)$$

α is a positive parameter, which scales the spread of sigma point distribution about its mean. μ is a normalization term, which offsets the effect of α . The α can be set sufficiently small that the higher order terms do not affect the filter. The SUT algorithm is as follows:

$$x_i^j = x_0 + \alpha(x_i - x_0)$$

$$W_i^j = W_0 / \alpha^2 + (1 - 1/\alpha^2) \quad i = 0$$

$$W_i^j = W_0 / \alpha^2 \quad i \neq 0$$

$$y'_i = f[\chi'_i]$$

$$\bar{y}' = \sum_{i=0}^{2n_x} W'_i y'_i$$

$$P'_i = \sum_{i=0}^{2n_x} W'_i \{y'_i - \bar{y}'\} \{y'_i - \bar{y}'\}^T + (1 + \beta - \alpha^2) \{y'_0 - \bar{y}'\} \{y'_0 - \bar{y}'\}^T$$

... (4.2)

where β is used for incorporating the higher order information.

4.2 The minimal skew simplex points:

The minimal skew sigma points are simplex set that is chosen to match the mean and covariance and minimize their skew. The minimum number of sigma points required to match the same mean and covariance of an n - dimensional random variable is $n+1$. These points have the important property that they can be constructed recursively. The set of points for the points in the $n+1$ dimension can be written in terms of those for the n th dimensional. The point selection algorithms is as follows:

1. Choose W_0

$$2. \text{ Choose weight sequence: } W_i = \begin{cases} \frac{1 - W_0}{2^n} & \text{for } i=1 \\ W_1 & \text{for } i=2 \\ 2^{i-1} W_1 & \text{for } i=3, \dots, n+1 \end{cases} \quad \dots(4.3)$$

3. Initialize vector sequence as:

$$\chi_0^1 = [0] \chi_1^1 = \left[-\frac{1}{\sqrt{2W_1}} \right] \text{ and } \chi_2^1 = \left[\frac{1}{\sqrt{2W_1}} \right]$$

4. Expand vector sequence for $j=2, \dots, n$ according to

$$\chi_i^{j+1} = \begin{cases} \begin{bmatrix} \chi_0^j \\ 0 \end{bmatrix} & \text{for } i=0 \\ \begin{bmatrix} \chi_i^j \\ -1 \\ \sqrt{2W_i^j} \end{bmatrix} & \text{for } i=1, \dots, j \\ \begin{bmatrix} 0_j \\ 1 \\ \sqrt{2W_j^j} \end{bmatrix} & \text{for } i=j+1 \end{cases} \quad \dots(4.4)$$

In this approach, the system dynamics are non-linear functions of the system states. Assuming again for simplicity, the case of a single moving target, the algorithm is as follows

4.3 Algorithm

1. To initialize the filter, calculate the update estimates of state vector and its covariance [17,18]
2. Find simplex sigma points (χ_i) and weights from the update estimates of the state and its covariance for each Montecarlo run. (where n is the size of the state vector) from eqns. (4.3,4.4.)
3. Find simplex sigma points (n_i) of acceleration process noise as in step 1 using zero mean and process noise covariance as P_x .
4. Scale these sigma points using eqn set (4.2).
5. Transform these sigma points through process model,

$$\chi_i(k+1/k) = f[\chi_i(k/k), u(k), n_i, k] \quad \dots(4.5)$$

and $W_l = W_l W_j$ where $l = 0..24$ ($6*4$). The f is given by eqn. (3.3)

6. Prediction of the state estimate at time $(k+1)$ with the measurement up to time k is given by,

$$\hat{X}(k+1/k) = \sum_{i=0}^{24} W_i \chi_i(k+1/k) \quad \dots(4.6)$$

7. The predicted covariance is given as

$$P(k+1) = \sum_{i=0}^{24} W_i [\chi_i(k+1/k) + \hat{X}(k+1/k)] [\chi_i(k+1/k) + \hat{X}(k+1/k)]^T \quad \dots(4.7)$$

8. Make a prediction in measurement as

$$\hat{Z}(k+1/k) = H \cdot \hat{X}(k+1/k) \quad \dots(4.8)$$

$$\text{Where } H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

9. Since the observation noise is additive and independent, the innovation covariance is given by

$$S(k+1) = R + H P(k+1/k) H^T \quad \dots(4.9)$$

where R is the observation noise vector covariance.

10. Input the observation and calculate innovation as

$$v(k+1) = Z(k+1) - \hat{Z}(k+1/k) \quad \dots(4.10)$$

11. Calculate the NIS as

$$NIS(k+1) = v(k+1)^T S(k+1)^{-1} v(k+1) \quad \dots(4.11)$$

12. Check if NIS of target has crossed the threshold of 79.5. If yes, scale process noise covariance and repeat steps (3-11) till NIS is brought below threshold.

13. Calculate Kaman Gain as

$$W(k+1) = P(k+1/k) H^T S^{-1}(k+1) \quad \dots(4.12)$$

14. Update of the error covariance is

$$P(k+1/k+1) = P(k+1/k) - W(k+1) S(k+1)^{-1} W(k+1)^T \quad \dots(4.13)$$

15. State update is

$$\hat{X}(k+1/k+1) = \hat{X}(k+1/k) + W(k+1) [Z(k+1) - \hat{Z}(k+1/k)] \quad \dots(4.14)$$

16. Go to step 2 to process the next observation sample.

Note that the above steps must be carried out for each of Monte Carlo run.

5. SIMULATION RESULTS

In order to verify the Unified Integrated Tracking Systems based on Unscented Kalman Filter proposed in this paper, computer simulation experiments were carried out and the results are presented in this section. Simulations were generated assuming additive white gaussian, mutually uncorrelated process and measurement noises. Also, the measurement noise vector is assumed to be mutually independent. Two sources having one multipath(coherent) radiate Radio waves and are received as narrowband planar wavefronts with center frequency $(f_c) = 0.4\text{GHz}$ by a LES array comprising 20 sensors. Sensor separation $\Delta = \lambda/2$. The signals were corrupted by spatially colored noise with an unknown covariance matrix. The figure 1 shows the DOA estimation for $\text{SNR} = -20\text{dB}$. 200 time samples are taken for each measurement. Initial DOAs are $\theta_1(t_0) = 27^\circ$ and $\theta_{1m}(t_0) = 40^\circ$, $\theta_2(t_0) = 36^\circ$ and $\theta_{2m}(t_0) = 65^\circ$. The DOAs are inherited from Extended Differential Music algorithm. DOA and range of target1 are 27 degrees 2236.06m respectively. For target2, they are 36 degrees and 282.84 m. The initial x and y velocities of target1 are 512 m/s and 616 m/s respectively. for target2, they are 512 m/s and 610 m/s. The process and measurement noise variances for both targets are 10^{-6} and $16 (\text{deg})^2$ respectively. Targets 1 and 2 maneuver at 30^{th} and 40^{th} samples respectively. The figures 3 and 4 shows AOA error plots of two targets using Unscented and Unscented Kalman Filter. From the error plots we can see that the UKF filter performance improved over Unscented Filter. The number of flops per sample is 71878 for UKF where as Unscented Filter is taking 134690 flops per sample i.e. nearly 50% computations are reduced. The number of memory locations required to implement the unscented filter for passive DOA tracking is 410 where as the reduced sigma point filter is taking 220 i.e. nearly 50% reduced.

6. CONCLUSIONS

In this paper, new algorithms are presented to adaptively estimate DOAs of multiple maneuvering targets, when LES array is used for making the measurements. The salient features of the proposed algorithm are (1) Noise subspace estimation is done by QR decomposition of the Difference matrix, which is formed from the data covariance matrix. Thus as compared to standard eigen-decomposition based methods which requires $O(M^3)$ computations where as proposed method requires $O(M^2)$ computations. (2) The proposed algorithm can able to discriminate between sources and their multipaths. The results obtained by this algorithm are then applied to a nonlinear state and Linear measurement Unscented filter and Unscented Kalman Filter to track targets (3) The algorithm uses simple NIS (Normalized innovation square) method to track maneuverable targets, which is very simple than previous methods[15,16]. The Reduced Sigma Point Unscented Filter (i.e. UKF) gives 50percent reduction in the computation and memory requirements over Unscented filter based tracker. Finally this proposed algorithm works in a realistic radar environment. Performance analysis of the proposed estimator and tracker is done for a typical realistic radar scenario by using computer simulation.

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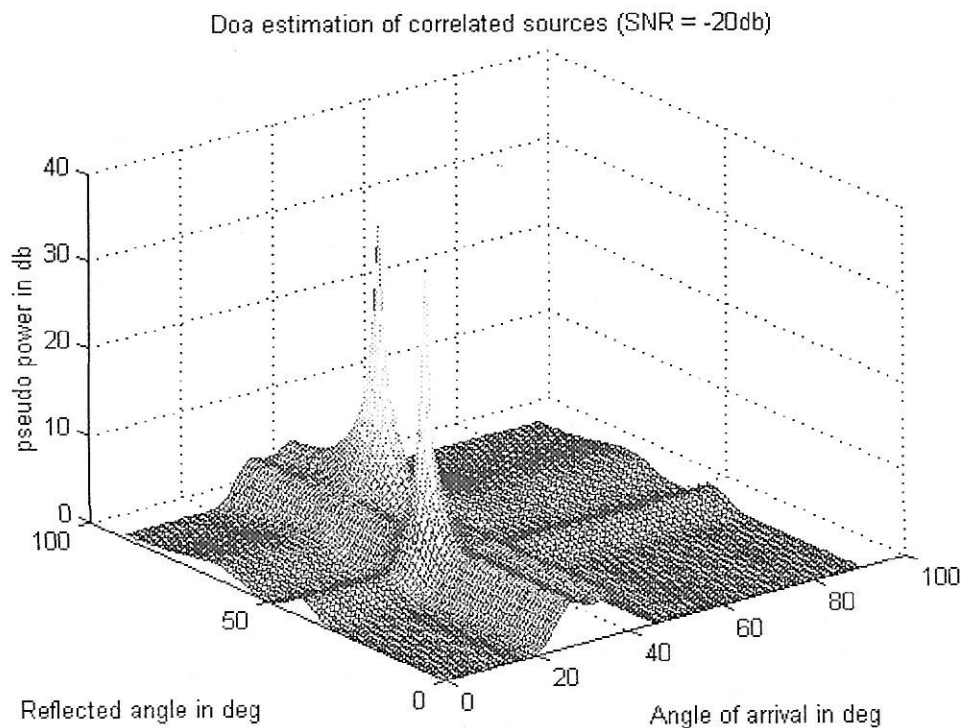


Fig 1: DOA estimation of correlated sources (SNR = -20dB)

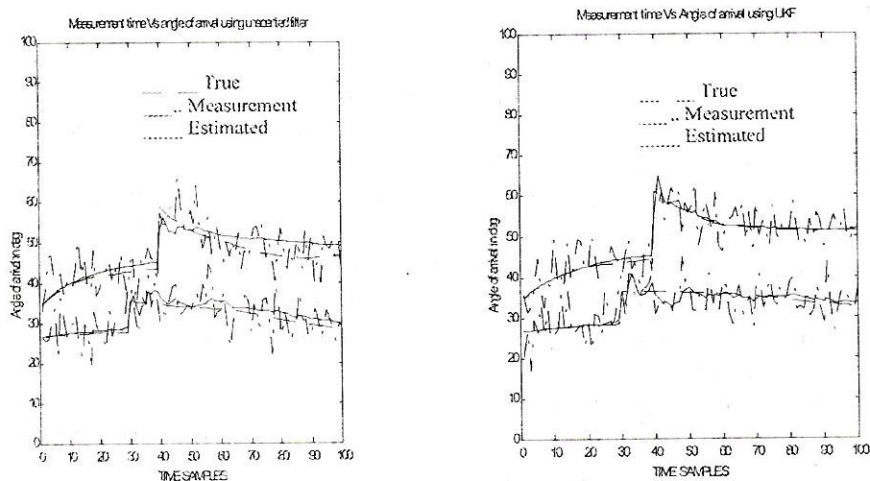


Fig 2: Passive DOA tracking of multiple maneuvering Targets Using Unscented Filter and UKF

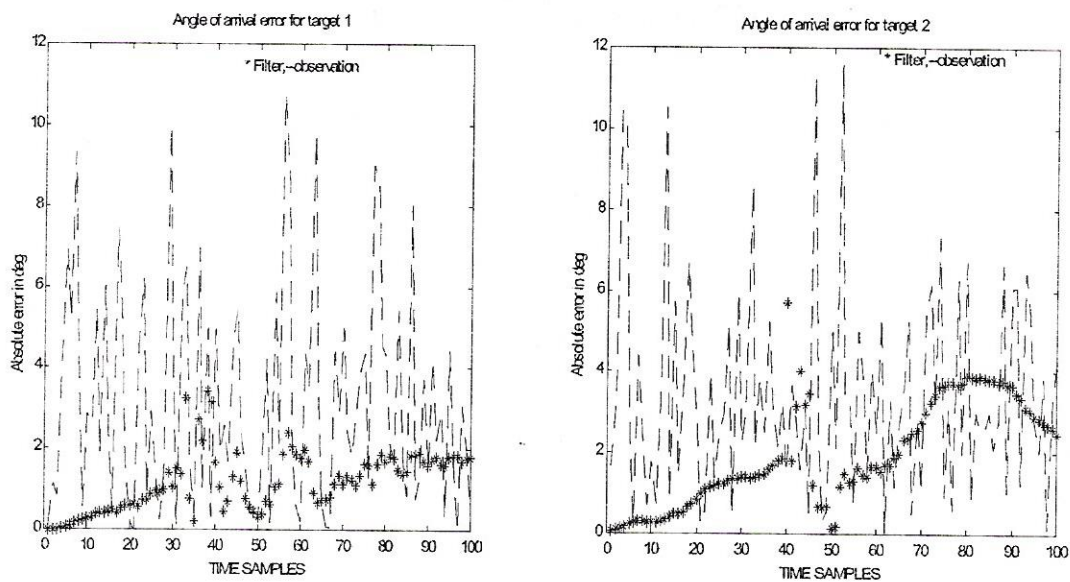


Fig 3: AOA error plots using Unscented Filter

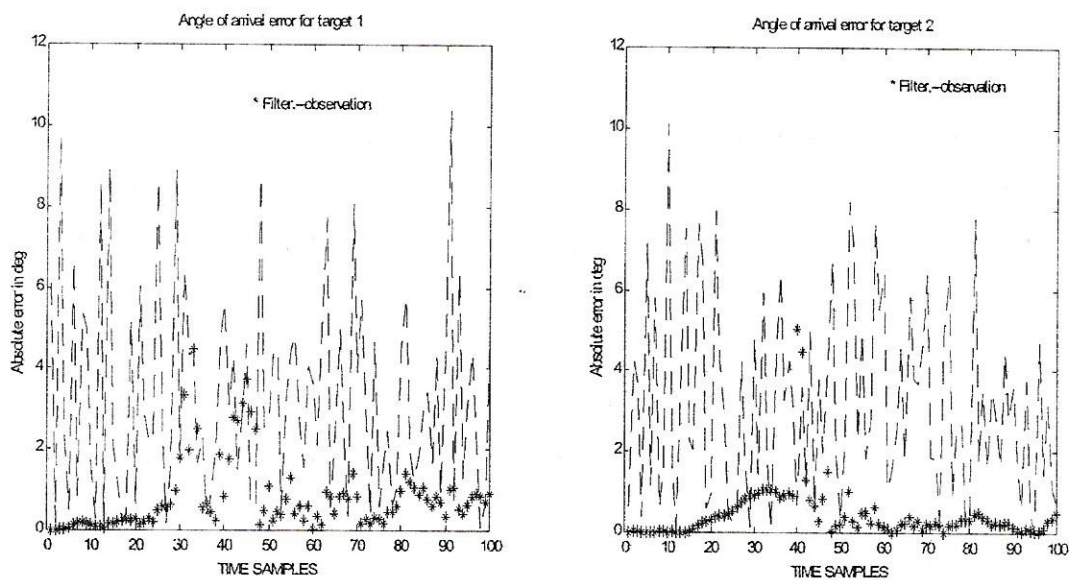


Fig 4: AOA error plots using UKF